SOFTWARE REQUIREMENTS SPECIFICATION

Live Map Events

User Application

Agile Edition · Sprint-Based Delivery

Version	1.0 — Living Document
Methodology	Agile / Scrum · 2-Week Sprints
Platform	iOS · Android · Web
Tech Stack	Flutter · Laravel 11 · Next.js (React)
Target Market	Syria
Date	March 2026

Living Document

This SRS is updated sprint by sprint. Sections marked [FUTURE] are high-level placeholders refined during sprint planning. Architecture and completed sprint specs are considered final once approved.

1. Introduction

1.1 Product Vision

Live Map Events is a location-based event and venue discovery platform for Syria. Users open the app to see a live interactive map populated with pins representing nearby events, restaurants, cafes, bars, and malls. They can explore events through a Tinder-style swiper, filter by category, and check in at locations.

1.2 Scope

This document covers the User Application only. The Business Owner Application (venue registration, event creation, analytics) will be documented in a separate SRS.

1.3 Sprint Roadmap

Sprint	Goal	Status
Sprint 1	Authentication (OTP via SMS + WhatsApp + Google) + Google Maps + Event Pins + Category Filters	In Progress
Sprint 2	Explore Feature — 5-Section Rule-Based Feed	Planne

1.4 Glossary

Term	Definition
PIN	Map marker representing a venue or event
PostGIS	PostgreSQL extension for native geospatial queries
ST_DWithin	PostGIS function — filters records within a given distance
OTP	One-Time Password delivered via SMS or WhatsApp
ALS	Alternating Least Squares — ML algorithm for recommendations (future)
FR / NFR	Functional / Non-Functional Requirement
TBD	To Be Defined — placeholder for future sprint

2. Technology Stack & Architecture

2.1 Stack Overview

Layer	Technology	Purpose
Mobile	Flutter 3.x	iOS + Android from one codebase
Web	Next.js 14 + React	Web app + marketing site + SEO
Backend API	Laravel 11	REST API shared by all clients
Database	PostgreSQL 15 + PostGIS	Relational DB with geospatial support
Cache	Redis 7	Feed caching, OTP storage, sessions
Auth	Firebase Auth + Laravel Sanctum	Google OAuth + API token management
Maps	Google Maps API	Interactive map + geocoding
Push	Firebase Cloud Messaging	Push notifications (iOS, Android, Web)
SMS / OTP	Twilio / WhatsApp Business API	OTP delivery via SMS or WhatsApp
Storage	AWS S3 / Cloudinary	Images: events, venues, avatars
DevOps	Docker + GitHub Actions	Containerized dev + CI/CD pipeline
Hosting	DigitalOcean / AWS	App server, managed DB + Redis

2.2 Architecture

Single Laravel monolith organized into domain Service classes (AuthService, GeoService, ExploreService, NotificationService). All clients share one REST API. The monolith is structured so that individual services can be extracted as microservices at scale (100k+ users) with minimal refactoring.

```
┌──────────────────────────────────────────────┐
│              CLIENT LAYER                    │
│  Flutter App    Next.js Web    Biz Owner (TBD)  │
└──────────────────────────────────────────────┘
                      │
                      ▼
┌──────────────────────────────────────────────┐
│          LARAVEL REST API  /api/v1           │
│  Auth · Explore · Pins · Events · Profile    │
│  Checkins · Notifications                    │
└──────────────────────────────────────────────┘
        │              │              │
        ▼              ▼              ▼
  PostgreSQL       Redis Cache     Firebase / S3
  + PostGIS        (feeds+OTP)     FCM / Twilio / WA
```

2.3 Project Structure

```
live-map-events/                  ← GitHub monorepo
├── mobile/                       ← Flutter
│   └── lib/
│       ├── core/                 ← API client, router, theme, auth
│       └── features/             ← auth, map, explore, swiper,
│                                    events, checkin, saved, notifications
```

```
├── web/                          ← Next.js
│   └── src/
│       ├── app/                  ← Pages (map, explore, events/[id])
│       ├── components/           ← Map, Explore, Auth, UI
│       └── lib/api.ts            ← Axios API client
├── backend/                      ← Laravel
│   └── app/
│       ├── Http/Controllers/Api/V1/
│       ├── Models/
│       └── Services/             ← AuthService, GeoService,
│                                    ExploreService, NotificationService
├── docs/                         ← SRS, API spec
├── docker-compose.yml
└── .github/workflows/ci-cd.yml
```

2.4 Database Schema

Table	Key Columns	Notes
users	id, phone, google_id, first_name, last_name, dob, avatar_url, profile_complete, fcm_token, location (PostGIS)	Phone or Google identity only — no password
user_interests	id, user_id, category	Multi-select tags from onboarding
venues	id, owner_id, name, type, address, city, location (PostGIS), is_active	Registered by business owners
events	id, venue_id, title, category, starts_at, ends_at, is_free, image_url	Created by venue owners
pins	id, venue_id, event_id?, type, location (PostGIS), has_promotion, label	One pin per active event or venue on map
checkins	id, user_id, venue_id, event_id?, checked_in_at	User check-in records
saved_events	id, user_id, event_id	Bookmarked / swiped-right events
user_interactions	id, user_id, event_id, interaction_type, duration_seconds, interacted_at	Phase 2 behavioral signals (future use)
device_tokens	id, user_id, token, platform	FCM push tokens

🗺 **PostGIS Strategy**

All location columns use GEOMETRY(Point, 4326) with GIST spatial indexes. This enables ST_DWithin (radius filter) and ST_Distance (ordering) directly in SQL — no application-level distance calculation needed.

3. Sprint 1 — Authentication & Map

Sprint 1 [In Progress]

OTP (SMS + WhatsApp) + Google Auth · Google Maps · Event Pins · Category Filters

Sprint 1 delivers the two foundational features: a secure authentication system and the core map experience with categorized pins. Both are required before any other feature can be built.

3.1 Authentication

Two sign-in methods only: Phone OTP and Google OAuth

- **OTP Flow —** User enters phone number (E.164 format, e.g. +963912345678) and selects delivery channel: SMS or WhatsApp. App sends to POST /auth/phone/request-otp with channel param. Backend generates 6-digit OTP, stores in Redis (5-min TTL), delivers via the chosen channel (SMS through Twilio, WhatsApp through WhatsApp Business API). User enters code → POST /auth/phone/verify-otp → returns Sanctum token + profile_complete flag.
- **Google Flow —** Flutter calls GoogleSignIn().signIn(), obtains ID token, sends to POST /auth/google. Backend verifies server-side with Google API, finds or creates user, returns token.
- **New users —** Redirected to profile setup (name, DOB, interests, location permission). Returning users go directly to the map.

Requirement	Priority
OTP is 6 digits, expires in 5 minutes, deleted after use	High
User selects OTP delivery channel: SMS or WhatsApp	High
WhatsApp OTP delivered via WhatsApp Business API (official template message)	High
SMS OTP delivered via Twilio as fallback or primary channel	High
Max 3 failed OTP attempts → 15-minute lockout	High
Max 3 OTP requests per phone per hour (rate limit)	High
Google ID token verified server-side — never trusted on client claim	High
Tokens stored in Flutter Secure Storage (Keychain / Keystore)	High
Profile setup: first name, last name, DOB (min age 16), interests (min 1), location permission	High
Session persists across app restarts	Medium

3.2 Google Maps + Event Pins

Map View

- Map centers on user GPS location. Default radius: 500m.
- If location denied: defaults to city center (Damascus).
- Standard gestures: pan, pinch-zoom, rotate.
- Re-fetches pins when user navigates > 100m from last fetch origin.

- Pin clustering applied when > 30 pins overlap at current zoom.

Pin Types & Category Labels

Pin Type	Icon / Color	Category Labels
restaurant	Red	food, brunch, tasting
cafe	Brown	coffee, music, coworking
bar / pub	Purple	nightlife, dj, happy hour
mall	Blue	shopping, sale, launch
general	Gray	tech, sports, arts, family, networking, outdoor, pop, rap, jazz, etc.

Category Filter Bar

- Horizontal scrollable pill bar at top of map screen.
- Tapping a category hides all non-matching pins in real time.
- Multiple filters combinable. 'All' clears all active filters.

Pin Detail Sheet

Tapping a pin opens a bottom sheet showing: venue name, category, event title (if applicable), date/time, distance from user, photos, promotion label, and a Check-in button.

3.3 Sprint 1 — API Endpoints

Method	Endpoint	Auth	Description
POST	/api/v1/auth/phone/request-otp	Public	Send OTP via SMS or WhatsApp (params: phone, channel: sms|whatsapp)
POST	/api/v1/auth/phone/verify-otp	Public	Verify OTP → return token
POST	/api/v1/auth/google	Public	Verify Google ID token → return token
POST	/api/v1/auth/logout	Bearer	Revoke current token
GET	/api/v1/auth/me	Bearer	Get current user + interests
GET	/api/v1/profile	Bearer	Get user profile

PUT	/api/v1/profile	Bearer	Update name, DOB
PUT	/api/v1/profile/interests	Bearer	Update interest tags
POST	/api/v1/profile/avatar	Bearer	Upload profile photo
GET	/api/v1/pins/nearby	Bearer	Map pins within radius (lat, lng, radius, types[]?, categories[]?)
GET	/api/v1/pins/{id}	Bearer	Pin detail — venue + event info
GET	/api/v1/events/{id}	Bearer	Full event detail
POST	/api/v1/checkins	Bearer	Record check-in (venue_id, event_id?)
GET	/api/v1/checkins/history	Bearer	User check-in history

4. Sprint 2 — Explore Feature

Sprint 2 [Planned]

5-Section Rule-Based Event Discovery Feed

The Explore screen presents events grouped into five independently ranked sections. Each section uses rule-based logic suited to its context. No machine learning in this sprint.

4.1 The Five Sections

Section	Definition	Ranking Logic
Tonight	Events starting today within radius	Time proximity — soonest start first (ASC starts_at)
Nearby	All active events within radius	Physical distance — PostGIS ST_Distance ASC
Interests	Events matching user's saved interest tags	Match score DESC, then distance ASC. Falls back to popularity if no interests set.
Trending	Highest check-in velocity within radius	Check-ins in last 2 hours DESC. Cached in Redis per city, refreshed every 5 min.
New	Events/pins created in last 7 days within radius	Newest first (DESC created_at). Gives new businesses immediate visibility.

4.2 Key Requirements

Requirement	Priority
All five sections loaded in a single API call	High
Each section supports independent pagination (page size: 10, max: 20)	High
Tonight: ranks by starts_at ASC; ongoing events (ends_at > NOW) included	High
Nearby: ranks using PostGIS ST_Distance ASC	High
Interests: matches user_interests tags; falls back to popularity for new users	High
Trending: check-in velocity cached in Redis (5-min TTL) per city	High
New: shows events with created_at within last 7 days, newest first	High
Empty sections hidden — not rendered as blank rows	Medium
Category filter applies to all sections simultaneously	High
Feed response cached in Redis: 3 min (personalized), 5 min (generic)	Medium
Search bar with keyword search (debounced 300ms) → GET /explore/search	High
Similar events shelf on pin detail: same category, same radius, max 4 results	Medium
API response: < 500ms cached, < 1.5s uncached	High

4.3 API Endpoints — Sprint 2

Method	Endpoint	Key Params	Description

GET	/api/v1/explore/feed	lat, lng, radius, category?, time_slot?, page_*?	Grouped 5-section explore feed
GET	/api/v1/explore/similar	event_id, lat, lng, radius	Similar events for detail shelf (max 4)
GET	/api/v1/explore/search	q, category?, date_range?, lat, lng	Keyword search — events + venues

5. Out of Scope

Feature	Notes
Email / password auth	OTP + Google only — by product decision
Facebook OAuth	Removed from requirements
In-app payments / ticketing	Future sprint
User-generated events	Business Owner App only
User-to-user messaging	Not in roadmap
Business Owner dashboard	Separate SRS + separate application

6. Definition of Done

A story is Done when: all acceptance criteria met, code reviewed, unit tests passing (min. 80% backend coverage), tested on iOS + Android, no new errors introduced, API documented, and this SRS updated with any scope changes.

— End of Document · Next update: before Sprint 3 planning —